

PROOF OF CONCEPT

Lot Genius

Azure Pre-Configuration Request

Prepared for: Steffes Group, Inc.

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Lot Genius PoC — Azure Pre-Configuration Request

To: Steffes Azure / Entra administrators **From:** Philippe Richard, Kadima Consulting —

philippe.richard@kadimaconsulting.com **Re:** Three one-time grants to unblock the Lot Genius PoC deployment

Kadima is deploying the Lot Genius PoC entirely via Terraform into a **new, isolated resource group** (`rg-steffes-lotgenius-poc`) in your subscription, reading your existing Synapse data **read-only**. Philippe already holds **Contributor at subscription scope** (standing, no PIM), so he can create all the infrastructure and run the database setup himself. **Three items remain that Contributor cannot self-serve** — they need a subscription Owner, the Synapse admin, and a directory admin respectively.

Do these three once — then we're unblocked

#	Task	Who must do it	Why it's needed
A	Grant Philippe Role Based Access Control Administrator (or User Access Administrator) on <code>rg-steffes-lotgenius-poc</code> and <code>rg-steffes-tfstate</code>	Subscription Owner	Terraform creates role assignments to wire the app's managed identity to AOAI / Storage / ACR / Key Vault. Contributor cannot create role assignments — <code>apply</code> fails here without this.
B	Run <code>GRANT SELECT</code> on the curated Synapse views in <code>sqldb-main</code> to the workload identity + Philippe	Synapse admin — Sean Todd	Supplies the authoritative "trusted numbers." This is a data-plane SQL grant Terraform/ARM cannot issue.
C	Admin-consent the Lot Genius bot app's Graph/Teams permissions (+ Teams app-catalog publish)	Global / Application Administrator	Required only for the Teams / M365 Copilot surface — a directory-level action separate from subscription RBAC. Needed at the publishing phase, not for infra/data.

Everything else — creating the Postgres / Foundry / Container Apps / Key Vault / storage, setting the Postgres Entra admin, running `CREATE EXTENSION vector` + schema, and the Terraform state storage — **Philippe handles directly under his existing Contributor role**.

A. Role-assignment rights (subscription Owner) — the one hard blocker

Contributor grants everything **except** `Microsoft.Authorization/roleAssignments/write`. The Terraform wires the workload managed identity to its data planes (Cognitive Services OpenAI User on the AI Services account, Storage Blob Data, AcrPull, Key Vault access), each of which is a *role assignment*. Without the right to create those, `terraform apply` stops at the IAM step.

Preferred (least-privilege): assign Philippe **Role Based Access Control Administrator** — scoped to just the two PoC resource groups: `rg-steffes-lotgenius-poc` and `rg-steffes-tfstate`. This role can *only* create role assignments (optionally constrained to specific roles), nothing else. **User Access Administrator** or **Owner** on those RGs also works if that is simpler on your side.

Fallback if no standing grant is possible: an Owner can instead create the ~5 managed-identity role assignments out-of-band after the identity exists; Kadima will supply the exact identity name and role/scope list. The standing grant above is cleaner and self-service.

Philippe's identity for the assignment:

`Philippe.Richard_kadimaconsulting.com#EXT#@steffesauctioneers.onmicrosoft.com` (object id `da0d8553-c290-42aa-a66a-223b25a0474c`).

B. Synapse GRANT SELECT — unblocks the "trusted numbers" (Sean Todd)

Lot Genius reads authoritative aggregates from the **existing** Synapse serverless system of record. **We create, modify, and migrate nothing in Synapse** — it is strictly a read source. The current Function already reads `sqlldb-main` over a managed identity, so a working read path exists; we extend the same pattern to the PoC workload identity and to Philippe for build-time verification.

Verified 2026-06-16 (Kadima): a connection from Philippe to `syn-nucleus-workspace-prod01-ondemand.sql.azuresynapse.net / sqlldb-main` using an Entra access token **passed the firewall and authenticated**, then stopped at `Login failed for user '<token-identified principal>' (SQL 18456)` — i.e. the **only** missing piece is the `CREATE USER ... FROM EXTERNAL PROVIDER + GRANT` below. Network, TLS, and token auth already work.

This is a **data-plane** grant — Azure RBAC / Terraform cannot issue it. The Synapse workspace Entra admin is `Sean.Todd@steffesgroup.com`, who can connect with Entra auth and run:

```
-- In database sqlldb-main, as the Synapse Entra/SQL admin:

-- Workload identity (the MCP server / embedding job) — exact name supplied by Kadima:
CREATE USER [lotgenius-poc-mi] FROM EXTERNAL PROVIDER;
GRANT SELECT ON [schema].[curated_view_1] TO [lotgenius-poc-mi];
GRANT SELECT ON [schema].[curated_view_2] TO [lotgenius-poc-mi];
-- (repeat per curated view; or GRANT SELECT ON SCHEMA::[curated] if a dedicated schema exists)

-- Philippe, for build-time verification (verified guest UPN):
CREATE USER [Philippe.Richard_kadimaconsulting.com#EXT#@steffesauctioneers.onmicrosoft.com] FROM EXTERNAL PROVIDER;
GRANT SELECT ON [schema].[curated_view_1] TO [Philippe.Richard_kadimaconsulting.com#EXT#@steffesauctioneers.onmicrosoft.com];
-- (repeat per curated view)
```

Read-only by design: **SELECT only** — no write, no DDL. `ALTER ROLE db_datareader ADD MEMBER [...]` achieves the same read-only scope across all objects if you prefer.

Firewall: the workspace uses an IP allowlist. Please add the operator / MCP **egress IP(s)** to the SQL firewall allowlist — Kadima will supply the exact address(es).

C. App registration + admin consent (directory admin) — publishing phase only

For the Teams / M365 Copilot surface, a **Global Administrator** or **Application Administrator** must (a) allow the Lot Genius bot **app registration** (or let Kadima create it and then grant **admin consent** for its Graph/Teams permissions), and (b) approve the **Teams app-catalog** publish. This is the only directory-level action and is **not** needed until we publish the appraiser-facing agent — it does not block the infrastructure or data work.

Environment specifics

Item	Value
Subscription	e1c620c2-e1a4-4ad7-b4bf-cb6746c41103 (Pay-As-You-Go)
Tenant	751d0f54-ed4d-4917-82f0-fac22a548e18
New PoC resource group	rg-steffes-lotgenius-poc (North Central US)
Terraform state RG	rg-steffes-tfstate
Postgres Entra admin group (Philippe is a member)	19b73e33-283a-4379-af76-ae6308b439a0
Synapse workspace (read-only)	syn-nucleus-workspace-prod01 / RG syn-nucleus-northcentralus-prod-01
Synapse endpoint / database	syn-nucleus-workspace-prod01-ondemand.sql.azuresynapse.net / sqlldb-main
Kadima contact	Philippe Richard — philippe.richard@kadimaconsulting.com

Thank you — once A and B are in place the PoC deploys end-to-end; C is needed only at the Teams-publishing step.